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Architecture Justification: Remote Team Time-Tracking & Project Approver System

Architecture Selection:

"We chose Client-Server Architecture for the Remote Team Time-Tracking & Project Approver System."

1. Architectural Choice:

We selected a **Client-Server Architecture** utilizing a web/desktop client layer interfacing with centralized backend services and persistent database storage via secure RESTful APIs.

2. Two Scenario-Related Reasons:

- **Geographical Distribution of Users:** Remote developers and engineering managers work asynchronously across disparate time zones and devices. A centralized server ensures all time logs, timers, and project assignments synchronize to a single source of truth in real time without conflicting timesheets.
- **Centralized Business Rules & Workflow Enforcement:** Approval policies (e.g., maximum daily hours, mandatory reviewer assignments, and project budget thresholds) are strictly enforced by centralized backend components (`Time Tracker` and `Approval Workflow`), preventing client-side tampering with recorded hours.

3. Security Advantage:

- **Role-Based Access Control (RBAC) & Data Encapsulation:** Remote developers are restricted to viewing and logging hours solely for their assigned tasks, while managers obtain elevated privileges for reviewing and approving team submissions. Centralizing data access behind API interfaces ensures sensitive wage, client billing, and operational data cannot be accessed directly by client-side inspection.

4. Performance Benefit:

- **Optimized Querying & Lightweight Clients:** Heavy analytical computations—such as calculating project burn rates, weekly timesheet aggregations, and automated approval checks—are handled on dedicated server hardware and optimized database queries. This keeps the client application lightweight and responsive even on low-bandwidth remote connections.